ECE 6580 – Module Project 5

Avalanche Photodiodes and Phototransistors: Two Optical Detection Technologies Beyond the Photodiode and Photoconductor

Blaine Swieder
Tennessee Technological University

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Abstract

Photodiodes and photo-conductors are foundational optical transducers; however, the broader landscape of light detection includes many other device classes with distinct advantages. In this paper, we examine two technologies that are not covered in Module 5. These are the avalanche photodiode (APD) and the phototransistor. For each device, the semiconductor physics, governing equations, key figures of merit, and representative applications are shown here with reference to the journal and conference literature. The Avalanche Photodiode (APD) achieves internal current gain of approximately 10–500× through impact ionization, hence enabling photon-level sensitivity in fiber-optic receivers, LiDAR, and single-photon counting. The phototransistor leverages bipolar transistor amplification (h_{FE} = 100–1000) to drive milliampere loads from moderate illumination without an external amplifier, finding widespread use in optocouplers and proximity sensors. Both devices are examined in the context of their gain-bandwidth-noise trade space and compared against the PIN photodiode baseline.

1 Introduction

The detection of light underpins fiber-optic communications, automotive LiDAR, biomedical imaging, and consumer electronics [1]. The PIN photodiode provides unity internal gain and excellent linearity, while the photoconductor offers simplicity but limited bandwidth and relatively high dark current. Two detector classes that overcome different limitations of these baselines are the APD and the phototransistor. They were chosen here since they represent fundamentally different gain mechanisms. More specifically, the field-driven impact ionization in the APD versus transistor current amplification in the phototransistor, and together they span applications ranging from photon-counting science to low-cost proximity sensing.

The APD is appealing when the photon flux is low and bandwidth must remain high. The phototransistor is attractive when simplicity, current gain, and low cost matter more than precision or speed. Both devices have been studied extensively in the primary literature and are analyzed below using governing equations, representative specifications, and application context drawn from published sources.

2 The Avalanche Photodiode (APD)

2.1 Physical Principle and Structure

An APD is a reverse-biased p-n junction operated near breakdown so that photogenerated carriers are accelerated to kinetic energies that are sufficient to ionize additional lattice atoms, which is done via a process known as *impact ionization* [2]. Each primary electron-hole pair can therefore trigger an avalanche of secondary pairs, producing internal current multiplication. The widely used separate-absorption and multiplication (SAM) structure keeps the high-field multiplication region physically distinct from the lower-field absorption, hence minimizing dark current while maximizing gain [3]. In a typical InGaAs/InP SAM-APD, an InGaAs absorption layer is combined with an InP multiplication region so that wavelength response and avalanche behavior can be optimized separately. For telecom wavelengths (1310 nm and 1550 nm), the absorption layer is usually InGaAs ($E_g \approx 0.75$ eV) and the multiplication layer is InP ($E_g \approx 1.34$ eV). For 905 nm LiDAR, silicon APDs are often preferred since their ionization-coefficient ratio gives lower excess noise [6].

2.2 Governing Equations

The ionizing rates for electrons (α) and holes (β) follows the Chynoweth model [2] given by:

$$\alpha(E) = A_e e^{-B_e/E}, \qquad \beta(E) = A_h e^{-B_h/E}. \tag{1}$$

Here A and B denote the material constants and E denotes the electric field. The ionization ratio $k = \beta/\alpha$ is approximately 0.02 for silicon and approximately 0.4 for InGaAs/InP [4]. For a uniform-field multiplication region of width W, the multiplication factor is

$$M = \frac{1-k}{\exp[-(1-k)\alpha W] - k}. \tag{2}$$

As bias approaches breakdown, $M \to \infty$. In practice, M is typically about 50–200 for silicon and 10–30 for InGaAs devices. The APD responsivity is given by

$$R_{\mathrm{APD}} = M R_0 = M \frac{\eta q \lambda}{hc}, \tag{3}$$

where η denotes the quantum efficiency. For silicon at 800 nm with $\eta = 0.85$ and $M = 50$, one obtains $R_0 \approx 0.55$ A/W and thus $R_{\mathrm{APD}} \approx 27.5$ A/W. Avalanche multiplication also introduces excess noise, commonly written in McIntyre form as [2]

$$F(M) = kM + \left(2 - \frac{1}{M}\right)(1 - k). \tag{4}$$

The corresponding APD noise-equivalent power can be expressed by the equation

$$\mathrm{NEP}_{\mathrm{APD}} = \frac{1}{R_0}\sqrt{2qI_{\mathrm{dark}}F(M)}. \tag{5}$$

The APD gain-bandwidth product is limited by avalanche transit time and is often approximated by [4]

$$\mathrm{GBP} \approx \frac{1}{2\pi k \tau_t M}, \tag{6}$$

which gives values on the order of 100–200 GHz for modern silicon and SAM-APD structures.

Table 1: Representative APD specifications (Si: Hamamatsu S12023; InGaAs: Hamamatsu G8931).

Parameter	Si APD	InGaAs APD
Wavelength range (nm)	400–1000	900–1700
Typical gain M	50–100	10–30
Responsivity at peak (A/W)	25–50	5–15
-3 dB bandwidth (MHz)	1000	800
Operating voltage (V)	100–150	25–75
Excess noise factor $F(M)$	≈ 4	≈ 7

2.3 Applications of the APD

APDs are utilized where internal gain is essential. In long-haul fiber-optic receivers, their gain is able to improve sensitivity by roughly 5–10 dB over a PIN front-end [4]. In single-photon counting for quantum key distribution and fluorescence microscopy, APDs are biased above breakdown in Geiger mode, where a single photon triggers a macroscopic current pulse that is then quenched electronically [5]. Automotive LiDAR systems use silicon APDs at 905 nm to achieve centimeter-scale range resolution at distances above 100 m [6]. In addition APD array are also used in PET scanners since they can replace photomultiplier tubes in MRI-compatible detector modules [7].

3 Phototransistor

3.1 The Physical Principle

A phototransistor is a bipolar junction transistor whose base-collector junction is exposed to light. Absorbed photons generate electron-hole pairs in the reverse-biased base-collector depletion region, and the resulting base current is amplified by the transistor action of the device. In the common floating-base configuration, the collector current is given by

$$I_C = h_{FE} I_{\text{photo}} + I_{CEO}, \tag{7}$$

where I_{CEO} denotes the dark current, and h_{FE} denotes the transistor current gain. Since h_{FE} may range from 100 to 1000, even a 1 μA photocurrent can produce 0.1–1 mA of collector current. This causes the phototransistor to be appealing in simple switching and isolation circuits, especially when the base terminal is left unconnected for maximum sensitivity [8, 9].

3.2 Governing Equations and Speed Limitations

A simple responsitivity estimate is given by

$$R_{\text{PT}} = h_{FE} \frac{\eta q \lambda}{hc}. \tag{8}$$

For silicon at 850 nm, with $\eta = 0.6$ and $h_{FE} = 300$, this gives $R_{\text{PT}} \approx 123$ A/W. In practice, this large apparent responsivity is bought at the cost of bandwidth and linearity.

Unlike the APD, the phototransistor speed is usually specified experimentally by rise time, fall time, or cutoff frequency rather than by a single universal closed-form expression. The response depends strongly on stored base charge, collector-base capacitance, load resistance, bias conditions, and whether the device is driven into saturation. For a first-order response, rise time and bandwidth are related approximately by

$$f_{-3\text{db}} \approx \frac{0.35}{t_r}. \tag{9}$$

Commercial silicon phototransistor optocouplers therefore tend to report microsecond-scale switching times and cutoff frequencies in the tens-of-kilohertz to low-megahertz range under stated test conditions. For example, the Vishay CNY17 reports a typical rise time of about 2 μs and a cut-off frequency of about 110 kHz in linear operation at $V_{CE} = 5$ V, $I_F = 10$ mA, and $R_L = 75\ \Omega$ [14, 15].

The low-frequency noise model is commonly described as the sum of shot noise and flicker noise:

$$S_{i,n}(f) = 2qI_C + \frac{K_f I_C^a}{f}, \tag{10}$$

where K_f and a denote the device constants. In phototransistor isolators, low-frequency burst noise and $1/f$ noise can be significant and are one reason these devices are poorly suited to precision low-level analog detection [10, 11]. The corresponding noise-equivalent irradiance is much worse than that of an APD, but still adequate for proximity sensing, switching, and optocoupling.

Table 2: Representative Silicon Phototransistor Specifications.

Parameter	Value	Conditions
Peak wavelength	570 nm	photopic response
Collector dark current I_{CEO}	< 100 nA	$V_{CE} = 5$ V
Collector current I_C	2.0 mA	1 mW/cm^2, 570 nm
Rise/fall time	15 μs each	$R_L = 1$ kΩ
Current gain h_{FE}	200–1000	varies with I_C

3.3 Applications of the Phototransistor

Phototransistors are widely used where low cost and simple interfacing matter more than ultimate sensitivity. In optocouplers, an LED-phototransistor pair provides a galvanic isolation between logic and power stages [8, 12]. In reflective and transmissive optointerrupters, a phototransistor detects the interruption of a light path and generates digital edges for encoder discs and position sensors [9]. Phototransistors are also common in proximity sensors and low-cost ambient-light sensing, and they can be integrated into BiCMOS optoelectronic circuits where their BJT-compatible process flow is convenient [9, 13].

4 A Comparative Analysis

Table 3 compares APDs and phototransistors against the PIN photodiode baseline.

Table 3: Comparative summary of PIN photodiodes, APDs, and phototransistors.

Metric	PIN photodiode	APD	Phototransistor
Internal gain	1	10–500	100–1000
Bandwidth	100 MHz–10 GHz	100 MHz–10 GHz	50 kHz–1 MHz
NEP (pW Hz$^{-1/2}$)	~ 1	< 0.1	~ 1000
Operating voltage	0–5 V	25–200 V	3–30 V
Linearity	Excellent	Good	Moderate
Cost / complexity	Low	Moderate–high	Very low
Typical use	Fiber-optic Rx	Lidar, photon counting	Optocoupler, proximity

The gain-bandwidth product captures the core tradeoff. At representative points, a PIN detector with unity gain and 10 GHz bandwidth has a GBP of around 10 GHz, an APD with $M = 100$ and a 1 GHz bandwidth has a GBP near 100 GHz, a phototransistor with $h_{FE} = 300$ and 0.5 MHz bandwidth has an effective gain-bandwidth product near 150 MHz. This large difference reflects

the advantage of avalanche multiplication over minority-carrier storage. The APD pays for its performance with a high-voltage bias supply and excess avalanche noise, whereas the phototransistor pays for its simplicity with low speed, higher noise, and poorer linearity. In short, the APD is a detector for difficult optical links, while the phototransistor is a detector for inexpensive switching and isolation tasks.

5 Conclusion

Two optical detection technologies beyond the PIN photodiode and photoconductor were examined. The APD achieves responsivities of roughly 5–50 A/W and sub-picowatt noise-equivalent power through avalanche multiplication that is governed by the McIntyre excess-noise framework. As a result, it is therefore the preferred detector for long-haul fiber links, automotive LiDAR, and photon-counting science. The phototransistor provides h_{FE}-amplified photocurrent in an extremely simple circuit, making it cost-effective for optocouplers, proximity sensors, and encoder discs despite its kilohertz-to-megahertz bandwidth limitation. In conclusion, both these two devices illustrate how semiconductor detector design can tailor the gain-bandwidth-noise tradeoff to applications that range from single-photon science to everyday consumer light sensing.

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